Sim2Real Ablation Matrix

Vision-based RL on a WidowX AI arm (Lift-Cube, Push-T). Each row is one option; the first four columns say how well it does, the last two what it costs. Solid border = measured, dashed = estimated. Factors sorted by how much their options differ on the real robot.

	SIM2REAL transfer	SIM SUCCESS	SAMPLE efficiency	ENCODING efficiency	WALL CLOCK cost	VRAM cost
HIGH SIM2REAL INFLUENCE						
CAMERA POSE CALIBRATION						
Calibrated, with small per-step pose and FOV jitter (current)	High	Mid	Mid	Mid	Low	Low
Calibrated, fixed	Mid	High	High	High	Low	Low
Uncalibrated (hand-placed, nominal FOV)	Low	High	High	High	Low	Low
CAMERA EXPOSURE AND WHITE BALANCE						
Matched to the training brightness	High	Mid	Mid	High	Low	Low
Camera defaults	Low	Mid	Mid	Low	Low	Low
ENCODER						
Local pretrained (DINOv2 + AFA)	High	High	High	High	Mid	Mid
Global pretrained (DINOv2 / R3M + linear)	Mid	Low	Low	Low	Mid	Low
Scratch (NatureCNN / CompactViT)	Low	Mid	Low	Mid	Low	Mid
BACKBONE TRAINING						
Frozen pretrained backbone (≈ 0.1 M trainable in AFA)	High	Mid	High	High	Mid	Mid
Fine-tuned pretrained backbone (≈ 21 M trainable)	High	High	Mid	High	High	High
Trained from scratch (≈ 0.1–1 M trainable)	Low	Mid	Low	Mid	Low	Mid
OBJECT AND GOAL COLOURS						
Narrow random range around the rig's colours	High	High	High	High	Low	Low
Random colours (full RGB)	High	Mid	Low	Low	Low	Low
Exactly fixed colours	Low	High	High	High	Low	Low
SMOOTHNESS PENALTIES						
Action-rate and near-goal speed penalty	High	Mid	Low	Mid	Low	Low
Weak or no smoothness penalty	Low	High	High	Mid	Low	Low
DYNAMICS MODEL						
Measured arm, object mass and friction	High	Mid	Mid	Mid	Low	Low
Default MJCF values	Low	High	High	Mid	Low	Low
LATENCY IN SIM						
Delay randomized around the measured latency	High	Mid	Mid	Mid	Low	Low
Fixed one-step delay	Mid	High	High	Mid	Low	Low
No delay	Low	High	High	Mid	Low	Low
LIGHTING RANDOMIZATION						
Matched to the rig with margin (current)	High	High	High	High	Low	Low
Wide random lighting (intensity, colour, direction)	High	Mid	Mid	Mid	Low	Low
Fixed lighting	Low	High	High	High	Low	Low
GOAL MARKER SHAPE						
Hollow outline	High	High	High	High	Low	Low
Filled T	Low	Mid	Mid	Low	Low	Low
MID SIM2REAL INFLUENCE						
ACTION TYPE						
Target-relative delta (target observed)	High	High	High	Mid	Low	Low
Measured-relative delta	Mid	Mid	Mid	Mid	Low	Low
Absolute joint targets	Mid	Low	Low	Mid	Low	Low
CONTACT PENALTIES						
Small penalty on pressing the T into the table	Mid	Mid	Mid	Mid	Low	Low
No contact penalties	Low	High	High	Mid	Low	Low
Penalty or termination on gripper–T force	Low	Low	Low	Mid	Low	Low
DYNAMICS RANDOMIZATION						
Measured model + randomized gains, damping, mass, friction	High	Mid	Low	Mid	Low	Low
Measured model, only friction randomized	Mid	High	High	Mid	Low	Low
TRAINING SCENE						
Texture randomization (1,203 textures)	High	Mid	Low	Low	Low	Low
Rig scene (one photo of the real table)	Mid	High	High	High	Low	Low
CONTROL RATE						
20 Hz	Mid	High	High	Mid	Low	Low
50 Hz	Low	Mid	Mid	Mid	Low	Low
GOAL SIGNAL						
Goal pose as input + marker	Mid	High	High	High	Low	Low
Marker in the image only	Low	Mid	Mid	Mid	Low	Low
PROPRIOCEPTION NOISE						
Noise on joint positions and velocities in training	Mid	Mid	Mid	Mid	Low	Low
Exact proprioception	Low	High	High	Mid	Low	Low
PPO SETTINGS						
Tuned (KL 0.05, entropy 0.001, std floor)	Mid	High	High	Mid	Low	Low
Loose (KL 0.2, no entropy, no floor)	Low	Mid	Mid	Low	Low	Low
LOW SIM2REAL INFLUENCE						
FEATURE-MAP RESOLUTION						
16 × 16 or 14 × 14 tokens (DINOv2 patch 14, R3M layer3)	Mid	High	High	High	Mid	High
7 × 7 cells (R3M layer4)	Mid	Mid	Low	Low	Low	Mid
INPUT IMAGE RESOLUTION						
224 × 224 px	Mid	High	High	High	Mid	Mid
128 × 128 px	Mid	Mid	Mid	Mid	Low	Low
LATENT SIZE (ENCODER OUTPUT)						
256 (current)	Mid	High	High	High	Low	Low
Large (1024)	Mid	High	Mid	Mid	Low	Low
Small (64)	Mid	Mid	Mid	Mid	Low	Low